

Question Paper

Exam Date & Time: 22-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Circuits and Systems [ECM_2123]

Marks: 50

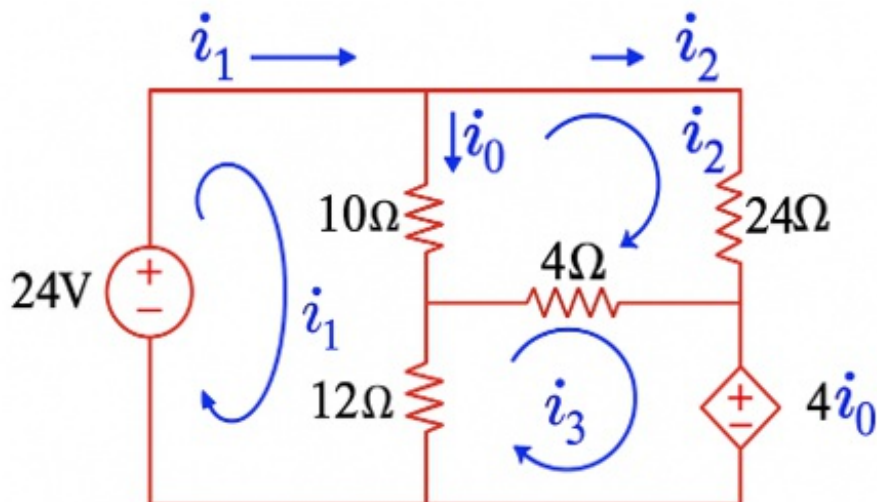
Duration: 180 mins.

PART-A

Answer all the questions.

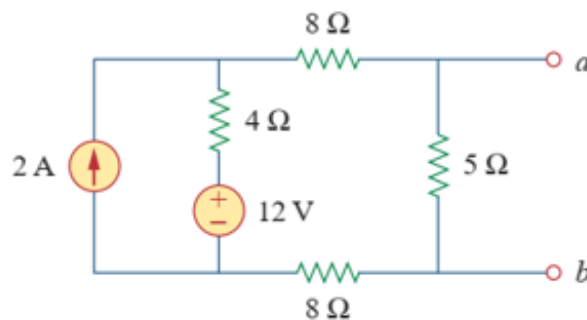
- 1) Use mesh analysis (Cramer's rule is optional) to find the current I_o in the circuit below. (4)

A)



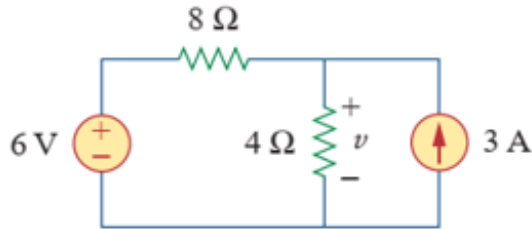
Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Find the Norton equivalent circuit of the circuit shown in the figure at terminals a-b. (4)



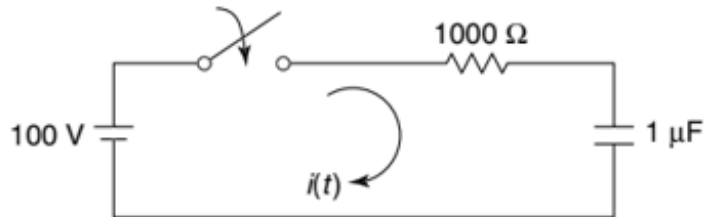
Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Use the superposition theorem to find 'v' in the circuit (2)



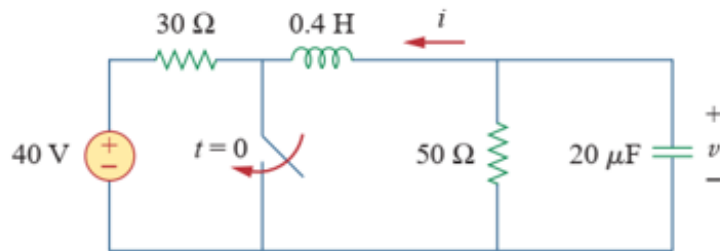
Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) In the given network, the switch is closed at $t=0$. With the capacitor uncharged, find value for i , $\frac{di}{dt}$, and $\frac{d^2i}{dt^2}$ at $t = 0^+$. (4)
- A)



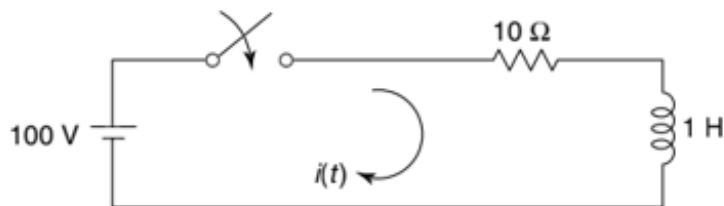
Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Find $v(t)$ for $t > 0$ in the given circuit. (4)



Course Outcome: "CO2" / Cognitive Level: "Application"

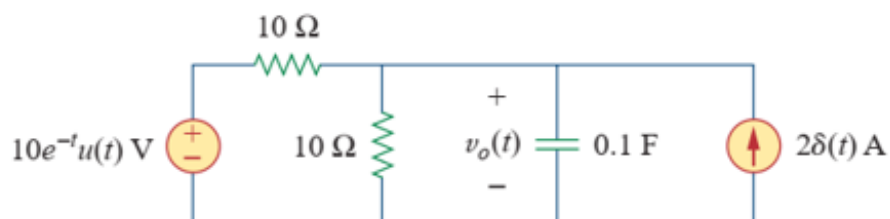
- C) In the given network, the switch is closed at $t=0$. With zero current in the inductor, find i , and $\frac{di}{dt}$ (2)



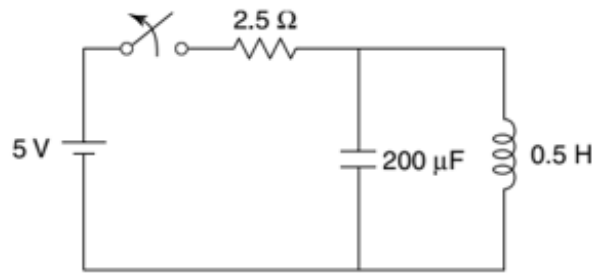
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Find $v_0(t)$ in the circuit. Assume $v_0(0) = 5V$. (4)

A)



- B) In the network of Figure, the switch is closed and a steady state is attained. At $t = 0$, the switch is opened. Determine the current through the inductor. (4)



- C) Find the Laplace transform of $\frac{1 - e^{-t}}{t}$. (2)

- 4) Examine whether the following signal is periodic or not? If periodic, determine the fundamental period. (4)

A) $3 \sin 200\pi t + 4 \cos 100t$

- B) Determine the power and rms value of the signal $x(t) = A \sin(\omega_0 t + \theta)$. (4)

- C) Sketch the following signals, $-2u(t + 2)$ (2)

- 5) Check the following systems are linear or not. $y(t) = x^2(t)$ (4)

- A) Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Determine whether the following system is linear, stable, causal, and time invariant using appropriate test: $y(n) = nx(n) + x(n + 2) + y(n - 2)$ (4)

- C) Find whether the following systems are dynamic or not. $y(n) = x^2(n)$ (2)

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECM_2123:Circuits and Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECM_2123:Circuits and Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)